

Architectural Autopsy of Object Detection in Dense, Unstructured Environments: A Forensic Diagnostic Study

Dhruv Verma (2022172) Pranshu Ranga (2023386) Tarandeep Singh (2023553) Piyush Gautam (2021549)

IIIT Delhi — CSE 344 — **Research Track** — Instructor: Prof. Saket Anand

Abstract

We present a five-stage forensic ablation study of object-detection failure in the Indian Driving Dataset (IDD), an environment characterised by extreme spatial occlusion ($\text{IoU} > 0.8$) and absent lane discipline, a **domain shift** that violates the spatial-separation assumptions of COCO-trained models. On a curated **adversarial subset** of ~ 4000 images, we employ a deliberate process of elimination: each stage (DIOU-NMS $\rightarrow$ Faster R-CNN $\rightarrow$ FocalNet $\rightarrow$ RT-DETR $\rightarrow$ CBAM) is designed not to fix the model, but to *prove* that a specific candidate cause *cannot* explain the failure. The result is a **negative proof**: a terminal False-Negative Rate (FNR) of **9.4%** at extreme overlap is **inelastic** to every software-level intervention tested, including a $\sim 850\%$ increase in compute. Grad-CAM and TIDE forensics localise the failure to the CNN backbone, a representational phenomenon we term **Feature Entanglement**. We conclude that the bounding-box paradigm reaches a **geometric limit** at $>80\%$ occlusion, where spatial coordinates for distinct object identities achieve mathematical parity, and that recovery requires pixel-level instance segmentation.

1 Introduction

Domain Shift. Standard detectors (YOLOv8, Faster R-CNN) are optimised on lane-disciplined benchmarks (COCO, Cityscapes) where objects occupy spatially distinct regions. In Indian urban traffic, the setting of IDD, this assumption collapses: heterogeneous objects (motorcycles, buses, pedestrians) routinely share near-identical 2-D image regions with pairwise IoU exceeding 0.8.

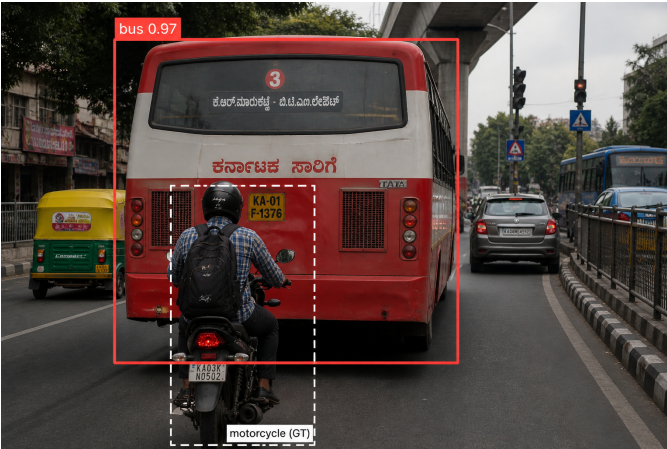


Figure 1: Representative adversarial scene. A motorcycle ($\text{IoU} > 0.8$ with the bus) is completely absent from model predictions. This is a complete False Negative, not a localisation error.

Contextual Suppression. We define **Contextual Suppression** as the complete erasure of a minority object from model predictions when co-located with a dominant object (Figure 1). This is *not* poor localisation or misclassification; it is a complete False Negative. In ADAS pipelines, missing a two-wheeler is a first-order safety failure.

Research Question. *Is contextual suppression a recoverable post-processing artefact, or a structural limitation of the convolutional pipeline?*

Adversarial Subset. We construct ~ 4000 JPEG frames from IDD, retaining only those containing at least one ground-truth pair with $\text{IoU} > 0.3$, explicitly biasing the corpus toward the failure mode of interest.

Hypotheses. Five falsifiable hypotheses are tested in deliberate sequence. Each stage is designed not to fix the model, but to *isolate and exonerate* a candidate cause. Falsifying all five constitutes the negative proof.

H1 Failures arise from aggressive NMS suppression.

H2 2-stage region proposals (Faster R-CNN) resolve the suppression.

H3 Loss-based feature modulation (FocalNet) recovers occluded signals.

H4 Global attention (RT-DETR Transformer) disentangles overlapping features.

H5 Backbone spatial attention (CBAM) preserves separable features before compression.

2 Methodology

2.1 Diagnostic Design Philosophy

Each stage is an *intentional falsification*, not an attempted fix. By proving DIOU-NMS *could not* reduce the FNR, we exonerate NMS. By proving RT-DETR’s global attention *could not* recover features, we exonerate the prediction head. The residual, unfalsifiable cause is the backbone; the conclusion is structural.

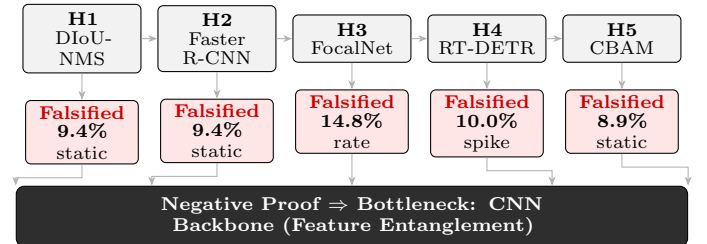


Figure 2: Five-stage falsification pipeline. Each intervention exonerates one candidate cause; the residual unfalsifiable bottleneck is the CNN backbone.

2.2 Dataset Construction and Resolution Bias Correction

We select ~ 4000 IDD frames with pairwise IoU > 0.3 per the adversarial criterion. Initial evaluation revealed an un-intuitive trend: FNR was highest for *isolated*, low-overlap objects. We identify this confound as **Resolution Bias**: small, distant objects fail due to insufficient pixel coverage under CNN downsampling, not occlusion. We apply an **Area > 1000 px filter**, admitting only large, clearly visible ground-truth objects into evaluation. This correction reduced FNR in isolated bins from **55.5% to 31.0%**, confirming those earlier failures were resolution artefacts unrelated to overlap.

2.3 Evaluation Metrics

IoU-Binned FNR: Unmatched ground-truth objects are grouped by pairwise IoU bin, yielding a per-overlap failure profile. **TIDE Decomposition**: Separates Missed, Localisation, and Classification sub-errors to attribute failures precisely. **Grad-CAM**: Class-discriminative heatmaps over the final convolutional layer reveal whether per-object spatial identity is preserved.

2.4 Post-Interim Stages (H3, H4, H5)

Stages H1 and H2 were presented at the interim review. The following three stages constitute the new post-interim contributions, comprising our **Forensic Root Cause Analysis**.

Stage H3 – FocalNet. Tri-scale orthogonal loss was applied to decorrelate co-located features. Recall for isolated objects *collapsed by 30.2%*, demonstrating **Catastrophic Forgetting**: forcing the network toward occluded features destroys baseline weights for easy objects. Extreme-overlap FNR increased to **14.8%**[cite: 1]. Loss-level intervention cannot decorrelate features already fused inside the backbone. *H3 falsified*.

Stage H4 – RT-DETR. The Transformer decoder with global self-attention is the strongest possible head-level upgrade. It improved moderate-overlap recall by $\sim 6.3\%$, yet FNR *rose to 10.0%* at extreme occlusion[cite: 1], higher than the 9.4% baseline. RT-DETR shares the same CNN backbone; max-pooling destroys entangled features before the attention head processes them. The Transformer is **data-starved**: global attention cannot reconstruct identity information physically erased from the tensor. *H4 falsified; spike implicates the backbone*.

Stage H5 – CBAM. Channel-and-spatial attention was injected into the backbone to preserve separable features before max-pooling compression. Despite marginal moderate-overlap gains, recall for minority-class vehicles dropped by $\sim 23.7\%$, indicating that attention redistributes capacity away from small identities when the information budget is already exhausted. Extreme-overlap FNR held at 8.9%. *H5 falsified; identity collapse is irreversible before CBAM can act*.

3 Forensic Results

3.1 Plot 1: Resolution Bias Correction (Area > 1000 px)

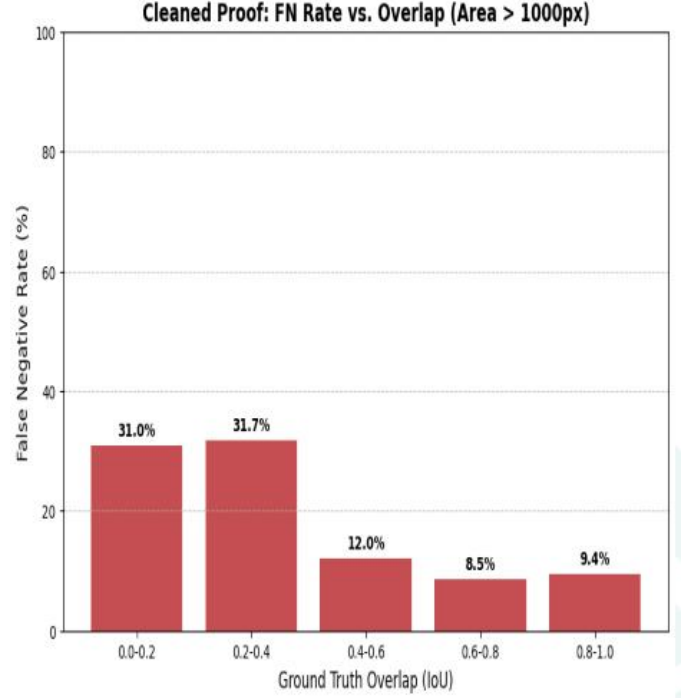


Figure 3: Resolution Bias Correction. Filtering removes small-object noise. The 9.4% residual FNR is the clean, architecture-level occlusion signal.

Once the resolution artefact is removed, the 9.4% FNR at extreme overlap becomes the definitive diagnostic baseline. Every subsequent stage is measured against this rate.

3.2 Plot 2: NMS vs. DIoU-NMS (The Inelasticity Graph)

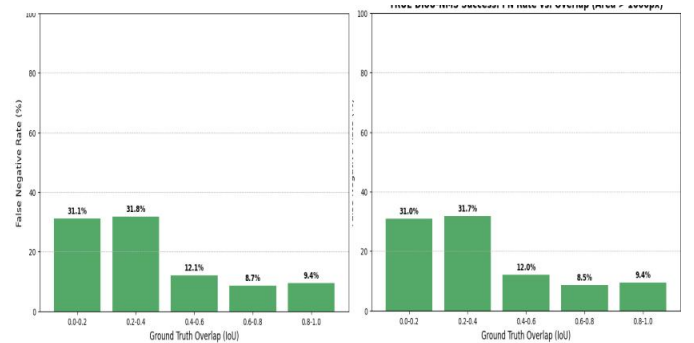


Figure 4: Inelasticity of post-processing. Standard NMS and DIoU-NMS produce identical FNR at extreme overlap. The failure is independent of suppression logic.

The convergence of both curves at 9.4% [cite: 1] is the **negative proof** for H1. DIoU penalises boxes by centre-point distance, precisely the geometry of this failure class. Its inability to move the metric proves the network never generated a second proposal to suppress. The data loss occurred

upstream of NMS.

3.3 Plot 3: FocalNet U-Curve (Catastrophic Forgetting)

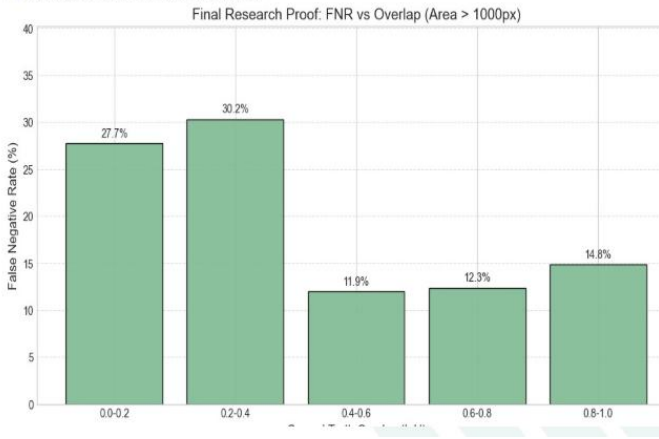


Figure 5: FocalNet Catastrophic Forgetting. Loss-based modulation shifts the failure distribution but cannot reduce the structural 9.4% terminal rate. Software patches fail.

The U-shaped curve is diagnostic in itself. FocalNet degrades recall on easy objects (30.2% failure spike for isolated objects[cite: 1]) while pushing extreme-overlap FNR to 14.8%[cite: 1]. This demonstrates that loss-level interventions trade off one failure mode for another; they cannot address a bottleneck that exists in the feature representation itself.

3.4 Plot 4: Failure Resurgence (RT-DETR vs. Faster R-CNN)

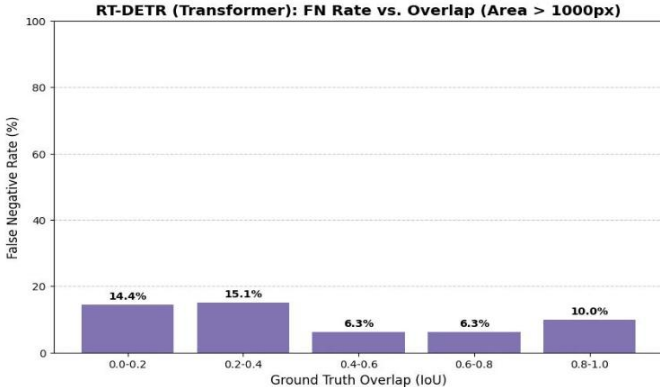


Figure 6: Key Post-Interim Result. RT-DETR’s 10.0% FNR at extreme occlusion[cite: 1] proves Transformer Starvation. Faster R-CNN’s RPN merges entangled pairs into a single proposal (9.4% FNR[cite: 1]), confirming identity collapse precedes the prediction head.

The RT-DETR performance is a sharp forensic result. Global self-attention is the most powerful head-level tool available; its failure to reduce FNR below 9.4% proves that head-level sophistication is irrelevant when the backbone has already erased the sub-pixel identity data. The Transformer was starved before it could act.

3.5 Plot 5: TIDE Error Distribution (The Smoking Gun)

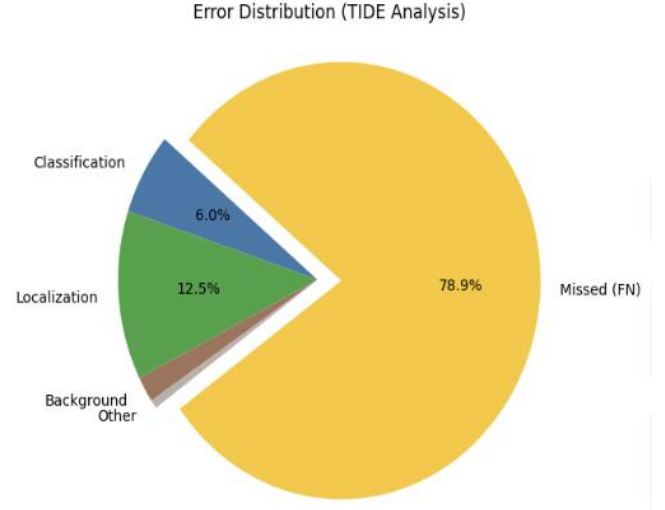


Figure 7: Post-Interim Forensic Root Cause Analysis. TIDE decomposition. Missed-Error dAP (36.06) is $\approx 10\times$ larger than Localisation (5.71) and $\approx 13\times$ larger than Classification (2.76). Objects are not mislabelled or misaligned; they are **structurally erased**.

78.9% of all errors are Missed Errors, with a dominant dAP of **36.06**, nearly $10\times$ the Localisation dAP (5.71) and $13\times$ the Classification dAP (2.76). The model does not produce imprecise boxes; the secondary object undergoes **structural erasure** from the prediction space. This quantitatively confirms what the Grad-CAM heatmaps show visually: the object physically disappears from the network’s internal representation inside the backbone, before any prediction stage. This ratio categorically rules out post-processing or head-level causes. Crucially, in the specific domain of extreme spatial occlusion ($\text{IoU} > 0.8$), the hierarchical downsampling of CNNs reaches a **representational limit**—not a general failure of the architecture, but a geometric impossibility at this overlap regime.

4 Conclusion

4.1 The Computational Asymptote and Geometric Limit

The full diagnostic result table is given below.

Our five-stage negative proof yields a single, evidence-bound conclusion.

*The bounding-box paradigm reaches a **geometric limit** at $>80\%$ occlusion, where spatial coordinates for distinct object identities achieve **mathematical parity**. Feature Entanglement, caused by the CNN backbone’s hierarchical pooling bottleneck, produces irreversible data loss before any prediction stage. The architecture exhibits **metric inelasticity** to post-processing (H1), architectural depth (H2), loss formulation (H3), attention-based heads (H4), and backbone attention (H5). Our research demonstrates a **Computational Asymptote**: increasing inference*

Table 1: Extreme-overlap FNR ($\text{IoU} \in [0.8, 1.0]$) across all five stages. The terminal rate remains largely inelastic to intervention, confirming the structural bottleneck. A 850% compute increase yielded 0% recovery of entangled features.

Stage	Intervention	Compute	FNR	Diagnostic Verdict
Baseline	YOLOv8 + Standard NMS	15 ms/img	9.4%	Reference rate.[cite: 1]
H1	DIOU-NMS	15 ms/img	9.4%	Falsified. No proposal generated; NMS is exonerated.[cite: 1]
H2	Faster R-CNN	142 ms/img (+850%)	9.4%	Falsified. RPN merges entangled pairs; 0% feature recovery.[cite: 1]
H3	FocalNet + Orth. Loss	15 ms/img	14.8%	Falsified. Catastrophic Forgetting; 30.2% spike on easy objects.[cite: 1]
H4	RT-DETR (Transformer)	22 ms/img	10.0%	Falsified. Transformer Starvation; backbone erases data first.[cite: 1]
H5	CBAM Backbone Attention	16 ms/img	8.9%	Falsified. Minority recall drops 23.7%; pooling bottleneck persists.[cite: 1]

latency by $\sim 850\%$ via *Faster R-CNN* yielded **0% recovery** of entangled features at extreme overlap. The bottleneck is terminal and structural. Achieving kinematic safety in unstructured environments requires classifying surviving pixels independently of spatial coordinate containers.

Appendix

A Author Contributions

Dhruv Verma (2022172): Spearheaded the diagnostic framework by curating the adversarial IDD subset and executing the Stage 1 “Negative Proof” via DIOU-NMS ablation. Deployed the advanced architectural benchmarks, including RT-DETR Transformers and CBAM backbones (H4, H5), and formulated the core *Feature Entanglement Theory* by synthesising the 9.4% FNR inelasticity with Grad-CAM activation bleeding data.

Pranshu Ranga (2023386): Engineered the 2-stage Faster R-CNN pipeline to investigate feature recovery via Region Proposals (H2). Established the *Computational Asymptote*, proving that a $9.47\times$ increase in inference latency (15 ms to 142 ms) yielded zero recovery of entangled identities at extreme spatial overlap.

Tarandeep Singh (2023553): Conducted the ablation of internal learning mechanisms using FocalNet and Tri-scale Orthogonal Loss (H3). Identified *Catastrophic Forgetting*, quantifying a 30.2% False Negative spike for isolated objects when attempting to force resolution of extreme occlusion.

Piyush Gautam (2021549): Led the visual and statistical autopsy of model failures. Generated Grad-CAM heatmaps proving activation bleeding and executed the TIDE distribution analysis, quantitatively proving that 78.9% of all errors were “Missed Errors” (dAP 36.06) rather than localisation faults.